MODULE 2 • DESCRIPTIVE STATISTICS

WHAT IS DESCRIPTIVE STATISTICS?

It is the part of statistics that helps us organise data, see patterns and explain what the data is telling us.

1 FREQUENCY DISTRIBUTION

Definition

A frequency distribution tells us how many times each value or category appears in the data. In simple words, it counts repeated values so patterns become easy to see.

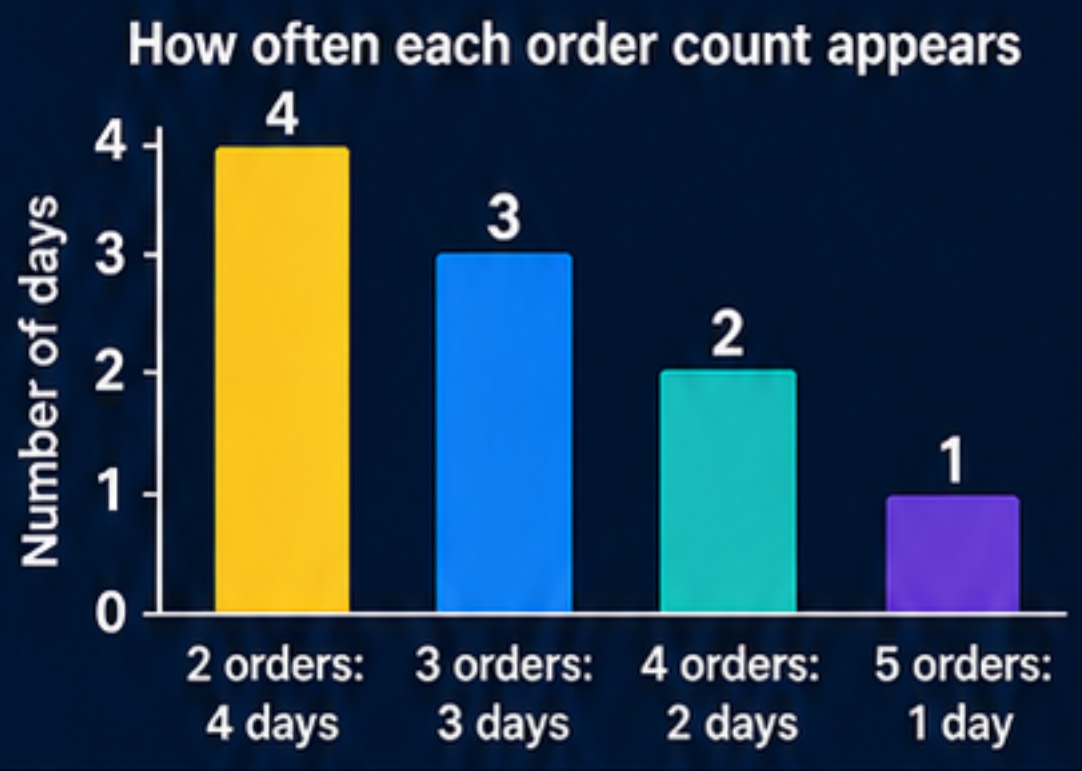
Why is it useful?

Raw data can look confusing. A frequency distribution groups the repeated values and shows which values occur often and which occur rarely.

EXAMPLE — Orders received in 10 days

Raw data:

2 3 2 4 3 2 5 3 2 4



BUSINESS USE CASE

An e-commerce team checks order frequency to understand usual demand. This helps them plan stock and schedule staff.

2 FREQUENCY TABLE

Definition

A frequency table is a neat table that writes each value in one column and the number of times it appears in another column.

Frequency table for daily orders

Orders	Frequency
2	4
3	3
4	2
5	1
Total	10

What can we understand?

2 orders appears most often.
It happened on 4 out of the 10 days.

Think like an analyst:

Which order count occurs most often?

MEAN (ARITHMETIC MEAN)

Definition

The arithmetic mean is found by adding all values in a dataset and dividing the total by the number of values. It is commonly called the average.

In simple words

The mean shares the total equally across all observations, giving us one value that represents the overall level of the data.

Formula

Mean = Sum of all values ÷ Number of values

$$\bar{x} = \Sigma x \div n$$

Example — Daily Orders

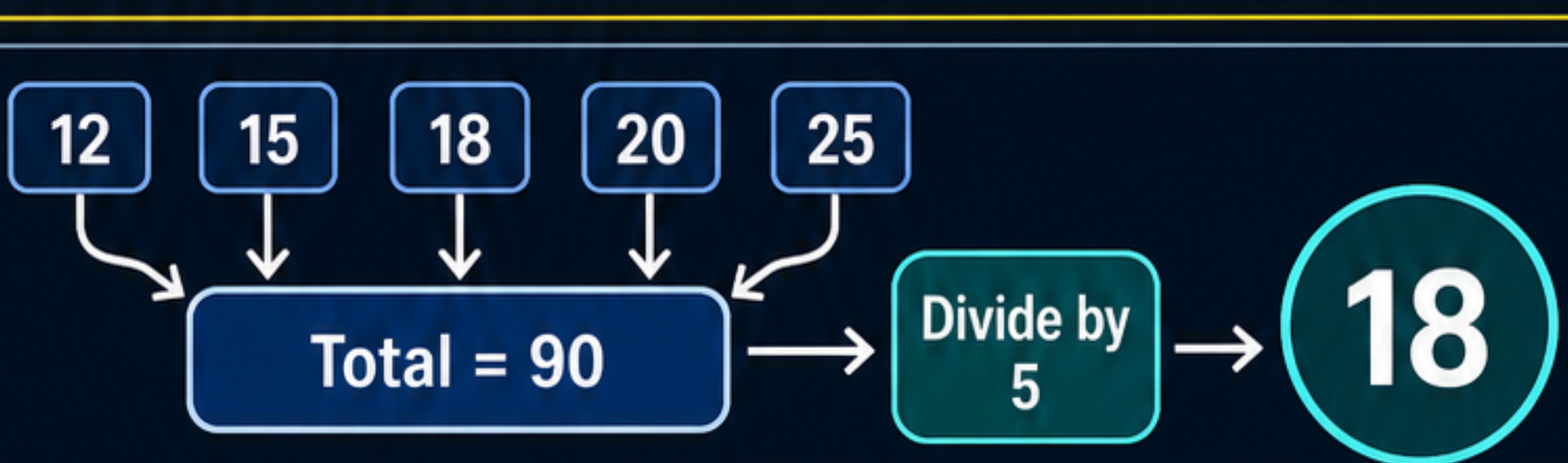
12	15	18	20	25

Step 1: Add the values = 12 + 15 + 18 + 20 + 25 = 90

Step 2: Count the values = 5

Step 3: Divide = 90 ÷ 5 = 18

Mean daily orders = 18



What does 18 mean?

Across these five days, the average number of orders was 18 per day. It does not mean every day had exactly 18 orders.

Business use case

A store manager uses mean daily orders to estimate normal demand and plan staff for a typical day.

Important note

A very high or very low value can change the mean because every value is included in the calculation.

Quick check: Total sales = ₹500 across 5 days. Mean daily sales = ?

₹100

WEIGHTED MEAN

Definition

A weighted mean is an average in which some values are given more importance than others. The importance assigned to each value is called its weight.

In simple words

When one value represents more customers, more sales or more units, it should influence the average more strongly. A weighted mean does this.

Formula

$$\text{Weighted mean} = \Sigma(\text{value} \times \text{weight}) \div \Sigma\text{weights}$$

How weights work

Larger weight → larger influence on the final average

Example — Product Ratings

Rating	Number of customers (weight)	Rating × weight
5	80	400
4	15	60
3	5	15
Total	100	475

$$\text{Weighted mean} = 475 \div 100$$

$$\text{Weighted mean rating} = 4.75$$

What does 4.75 mean?

The overall rating is 4.75 because the 80 customers giving 5 stars have the greatest influence.

Business use case

Online stores use weighted mean ratings so products with many reviews have more influence than products with only a few reviews.

Important note

Use a weighted mean only when weights represent a real and meaningful difference in importance.

A score of 90 has weight 3 and a score of 60 has weight 1. Weighted mean = ?

82.5

MEDIAN

Definition

The median is the middle value in a dataset after the values are arranged from smallest to largest. It divides the ordered data into two equal halves.

In simple words

Put the values in order and find the value in the centre.
Half the values will be below it and half will be above it.

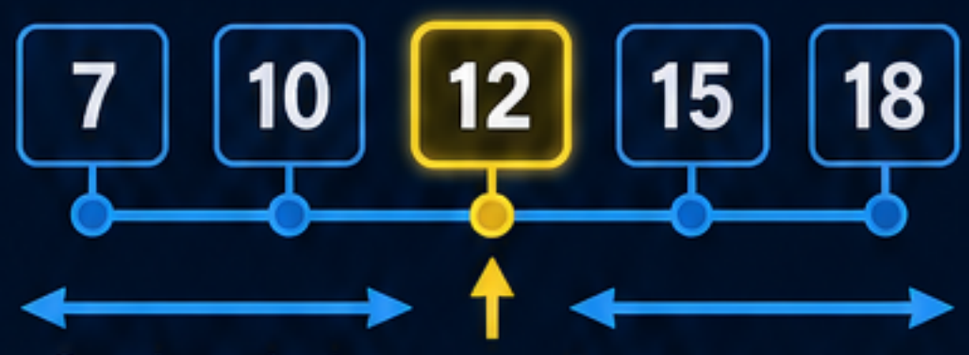
First rule

Always sort the data before finding the median.

1 Example 1 — Odd number of values

Unsorted row: 15 | 7 | 12 | 18 | 10

Sorted row: 7 | 10 | **12** | 15 | 18



There are 5 values. The middle is the 3rd value.

Median = 12

2 Example 2 — Even number of values

Sorted row: 5 | 8 | 11 | 14

There are 4 values. The two middle values are 8 and 11.

Median = (8 + 11) ÷ 2 = 9.5

Why is the median useful?

It shows a typical middle value and is less affected by extremely high or low values.

Business use case

A company can use median salary to describe typical pay when a few executive salaries are much higher than most salaries.

Quick check

Find the median: 4, 9, 2, 7, 5

5

MODE

Definition

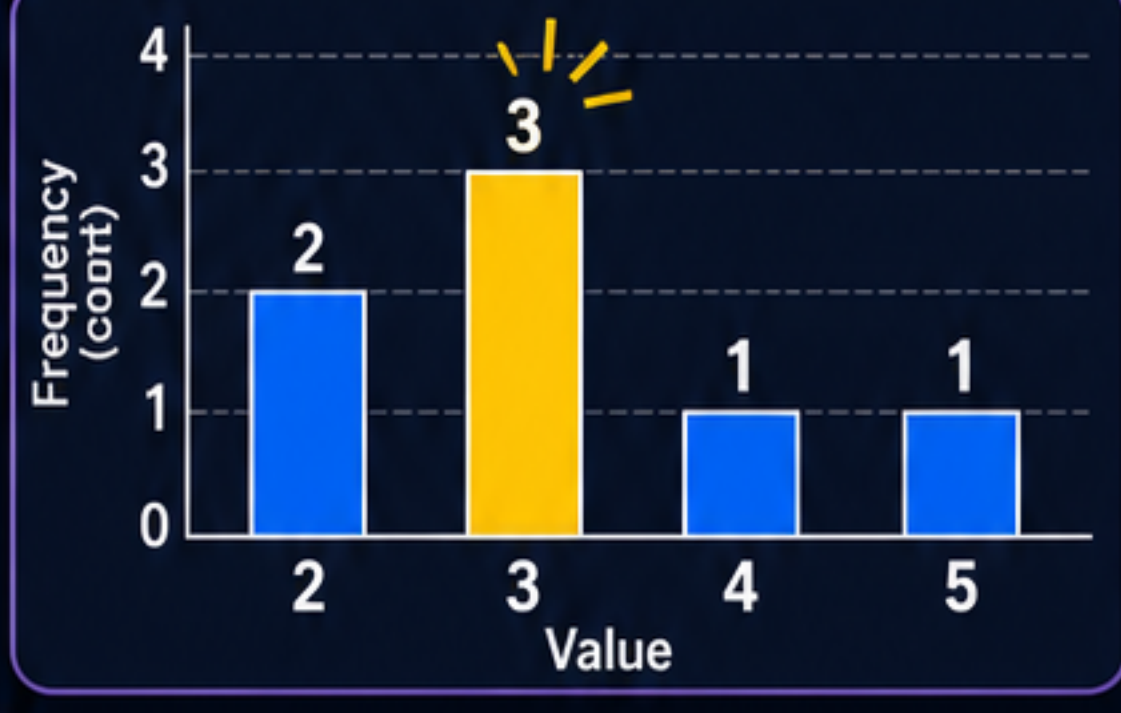
The mode is the value or category that appears most often in a dataset. A dataset can have one mode, more than one mode, or no mode.

In simple words

Find the value that occurs the most times. That most frequent value is the mode.

1 Example 1 — One mode

Data	2	3	3	4	3	5	2
Frequency	2→2 times	3→3 times	4→1 time	5→1 time			



Mode = 3

2 Example 2 — Two modes

Data	Red	Blue	Red	Green	Blue

Red and Blue each appear 2 times. This dataset has two modes.

Modes = Red and Blue

Possible outcomes

One mode – One value appears most often

More than one mode – Two or more values tie for most frequent

No mode – Every value appears the same number of times

Why is mode useful?

Mode is especially useful for category data, where adding or averaging labels would not make sense.

Business use case

A retailer uses mode to identify the most frequently sold product size or the most common payment method.

Quick check: Find the mode:
Small, Medium, Small, Large, Small, Medium

Small

GEOMETRIC MEAN

Definition

The geometric mean is an average calculated by multiplying all positive values together and then taking the root based on how many values there are.

In simple words

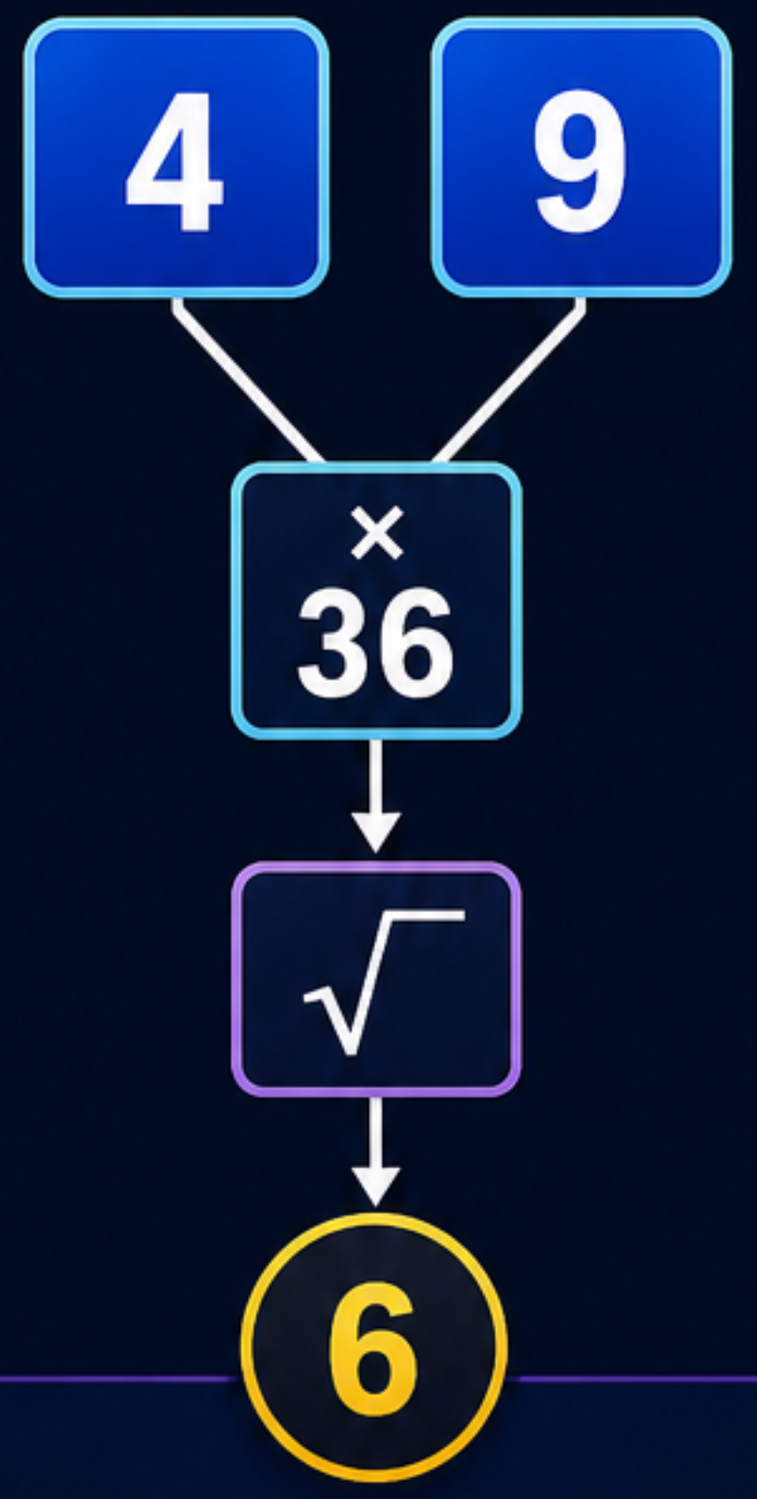
It is useful when values grow or shrink by percentages over time, because it reflects the combined or compounded effect.

Formula

Geometric mean = $\sqrt[n]{(x_1 \times x_2 \times \ldots \times x_n)}$

Important rule

Geometric mean is used only with positive values. Zero and negative values are not suitable.



Example — Two values

Values: 4 and 9

Step 1: Multiply = $4 \times 9 = 36$

Step 2: Take the square root = $\sqrt{36} = 6$

Geometric mean = 6

Growth example — Sales growth

Year 1 growth = 10% Year 2 growth = 21%

Growth factors: 1.10 and 1.21

Geometric mean factor = $\sqrt[2]{(1.10 \times 1.21)} = 1.154$

Average compound growth ≈ 15.4%

Why use it?

It gives a more realistic average for repeated percentage changes than simply adding the percentages and dividing.

Business use case

Analysts use geometric mean to summarise annual investment returns, sales growth rates and website traffic growth over several periods.

Quick check | Find the geometric mean of 1 and 16

4

HARMONIC MEAN

Definition

The harmonic mean is an average calculated by dividing the number of values by the sum of their reciprocals. It is mainly used for rates and ratios.

In simple words

When the same distance or the same amount of work is completed at different rates, the harmonic mean gives the correct average rate.

Formula

Harmonic mean = $n \div (1/x_1 + 1/x_2 + \ldots + 1/x_n)$

When should you use it?

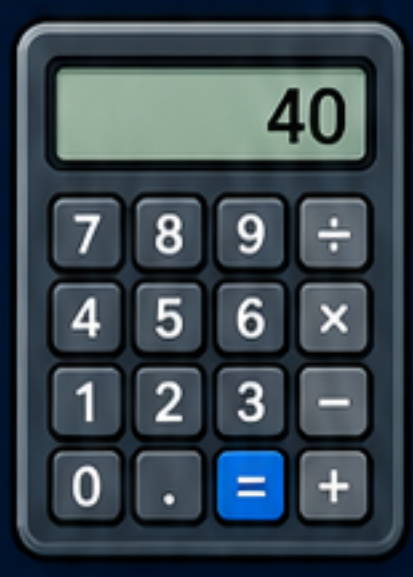
Use it for rates such as speed, cost per unit, productivity rate or price-to-earnings ratio when each part has equal weight.

Example — Average speed

A car travels the same distance twice: 60 km/h on the first trip and 30 km/h on the return trip.

Harmonic mean = 2 ÷ (1/60 + 1/30)

Harmonic mean = 2 ÷ (3/60) = 40



40 km/h

Average speed = 40 km/h

Why not simply use 45?

The car spends more time travelling at 30 km/h, so the slower speed has a bigger effect on the overall journey.

Business use case

A delivery business can use harmonic mean to calculate a realistic average delivery speed across routes of equal distance.

Important note

All values must be positive, and the quantities being averaged should be rates measured over equal units.

Quick check strip

Same distance at 20 km/h and 60 km/h.
Average speed = ?

30 km/h

3 MEASURES OF POSITION

Measures of position tell us where a particular value lies in a dataset. They help us understand the relative standing or rank of data points.

1 PERCENTILES

Definition

The p^{th} percentile is the value below which p% of the data falls.

For example, the 75^{th} percentile (P75) means 75% of values are less than or equal to this value.

Example – Exam Scores

Scores (sorted):

Business use case

In sales analysis, the 90^{th} percentile of order value shows the threshold below which 90% of orders fall. It helps in identifying top-value customers.

2 QUARTILES

Definition

Quartiles divide the data into four equal parts.

- Q1 (25th percentile)
- Q2 (50th percentile) = Median
- Q3 (75th percentile)

Example – Monthly Salaries (sorted)

Salaries (₹):

Business use case

HR teams use quartiles to compare employee salaries. Q1 shows lower earners, Q2 is the middle point, and Q3 shows higher earners.

3 INTERQUARTILE RANGE (IQR)

Definition

IQR is the range between the first quartile (Q1) and the third quartile (Q3).

IQR = Q3 – Q1

Example – Using the salary data above

Q1 = 25k, Q3 = 55k

IQR = 55k – 25k = 30k

Why useful?

IQR shows the spread of the middle 50% of data and is not affected by extreme values (outliers).

Quick check

Data (sorted): 5, 12, 18, 20, 25, 30, 40, 50

Find Q1, Q2 (Median), Q3 and IQR.

Answer

Q1 = 15, Q2 = 22.5,
Q3 = 35, IQR = 20

Key takeaway

Percentiles, quartiles and IQR help us understand the position and spread of data—not just the center.

Real world impact

These measures are widely used in finance, HR, marketing, quality control and performance analysis.

4 RANGE

Definition

Range is the difference between the highest (maximum) value and the lowest (minimum) value in a dataset.

Formula

Range = Maximum – Minimum

Example – Delivery times (in minutes)

Delivery times (minutes):

18 | 22 | 27 | 30 | 35 | 40 | 45

Maximum = 45 min

Minimum = 18 min

Range = 45 – 18

27 minutes

Business use case

A logistics company uses range to quickly understand how much the delivery time varies in a day.

Important note

Range is easy to calculate but can be highly affected by extreme values (outliers).

5 VARIANCE (Measure of Dispersion)

Definition

Variance measures how much the data values differ from the mean. It is the average of the squared differences from the mean.

Formula

Population Variance $(\sigma^2) = \frac{\Sigma (x_i - \mu)^2}{N}$

Sample Variance $(s^2) = \frac{\Sigma (x_i - \bar{x})^2}{(n - 1}$

Example – Number of calls received in a call center

Data: 2, 4, 6, 8, 10

Step 1: Find mean $(\bar{x})$

Mean = (2+4+6+8+10)/5 = 30/5 = 6

Step 2: Find $(x_i - \bar{x})^2$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	2−6 = −4	16
4	4−6 = −2	4
6	6−6 = 0	0
8	8−6 = 2	4
10	10−6 = 4	16
Sum		40

Step 3: Variance

Sample variance (s^2)

= 40 / (5−1)

= 40 / 4 = 10

Variance = 10

Business use case

A call center manager uses variance to understand the consistency in the number of calls. A high variance means call volume is unpredictable; a low variance means it is stable.

Key takeaway

Variance is always positive. It is measured in squared units.

6 STANDARD DEVIATION (Measure of Dispersion)

Definition

Standard deviation is the square root of variance. It tells us how much the data typically varies from the mean, in the original unit.

Formula

Population SD $(\sigma) = \sqrt{\sigma^2}$

Sample SD $(s) = \sqrt{s^2}$

From the previous example

Variance $(s^2) = 10$

Standard Deviation $(s) = \sqrt{10}$

$s \approx 3.16$

Business use case

A sales analyst uses standard deviation to measure consistency in daily sales. Lower SD means sales are consistent; higher SD means sales are more variable.

Interpretation guide

- SD = 0 → All values are same
- Small SD → Data points are close to mean
- Large SD → Data points are far from mean

Quick check

Data: 5, 5, 5, 5, 5

Variance = 0, SD = ?

Answer: 0

> Variance and Standard Deviation are the most important tools to understand data spread. They help us measure risk, variation and uncertainty in real-world data.

7 MEAN ABSOLUTE DEVIATION (MAD)

Definition

Mean Absolute Deviation (MAD) is the average of the absolute differences between each value and the mean. It shows the average deviation without squaring the values.

Formula

$$MAD = \frac{\sum |x_i - \bar{x}|}{n}$$

Example – Daily sales (in ₹)

Data:

10	12	14	16	18

Step 1: Find mean ($\bar{x}$)

Mean = (10+12+14+16+18)/5 = 70/5 = **14**

Step 2: Find $|x_i - \bar{x}|$

x_i	$x_i - \bar{x}$	$\lvert x_i - \bar{x} \rvert$
10	10 – 14 = –4	4
12	12 – 14 = –2	2
14	14 – 14 = 0	0
16	16 – 14 = 2	2
18	18 – 14 = 4	4
Sum		**12**

Step 3: Find MAD

$$MAD = \frac{\text{Sum of } |x_i - \bar{x}|}{n}$$

MAD = 12/5 = **2.4**

Interpretation

On average, the daily sales deviate from the mean (₹14) by ₹2.4.

Business use case

Retail businesses use MAD to understand the average variation in daily sales from the average. It helps in demand forecasting and inventory planning.

Why useful?

MAD is easy to understand because it uses absolute values. It is less affected by extreme values than variance but not as widely used as standard deviation.

8 COEFFICIENT OF VARIATION (CV)

Definition

Coefficient of Variation (CV) is the ratio of standard deviation to the mean, expressed as a percentage. It compares the relative variation between datasets with different units or scales.

Formula

$$CV = \left(\frac{\text{Standard Deviation}}{\text{Mean}}\right) \times 100$$

Example – Comparing two products

Product	Mean Sales (₹)	SD (₹)	CV Calculation	CV (%)
A	1000	150	(150/1000) × 100	**15%**
B	3000	300	(300/3000) × 100	**10%**

Interpretation

Product A has more relative variation (15%) compared to Product B (10%). So, sales of Product A are more inconsistent.

Business use case

Companies use CV to compare consistency between products, markets or teams. Lower CV means more consistency.

Key takeaway

Use CV when comparing datasets with different means or units (e.g., sales in ₹ vs quantity in units).

Quick check

Mean = 20 and Standard Deviation = 4.
What is the Coefficient of Variation?

20%

9 PERCENTILES

Measures of Position

Definition

The p^{th} percentile is the value below which p% of the observations fall.

Example: 75th percentile (P75) means 75% of values are less than or equal to this value.

In Simple Words

Percentiles divide the dataset into 100 equal parts.

It helps us understand the relative standing of a value.

Uses / Why Useful?

- Identifies performance levels (e.g., top performers)
- Helps in benchmarking
- Useful in risk analysis
- Common in exams, sales, finance, quality control

EXAMPLE – Exam Scores

Data (already sorted):

10	20	30	40	50	60	70	80	90	100

Formula (for ungrouped data)

$$\text{Position of } P_p = \frac{p}{100} \times (n+1)$$

where, p = percentile (e.g., 75 for P75)

n = number of observations

(n+1) gives the exact position

Find P75 (75th Percentile)

Step 1: Count the observations

n = 10

Step 2: Find the position

Position of P75 = $\frac{75}{100} \times (10+1)$

$= 0.75 \times 11$

$= 8.25^{th}$ position

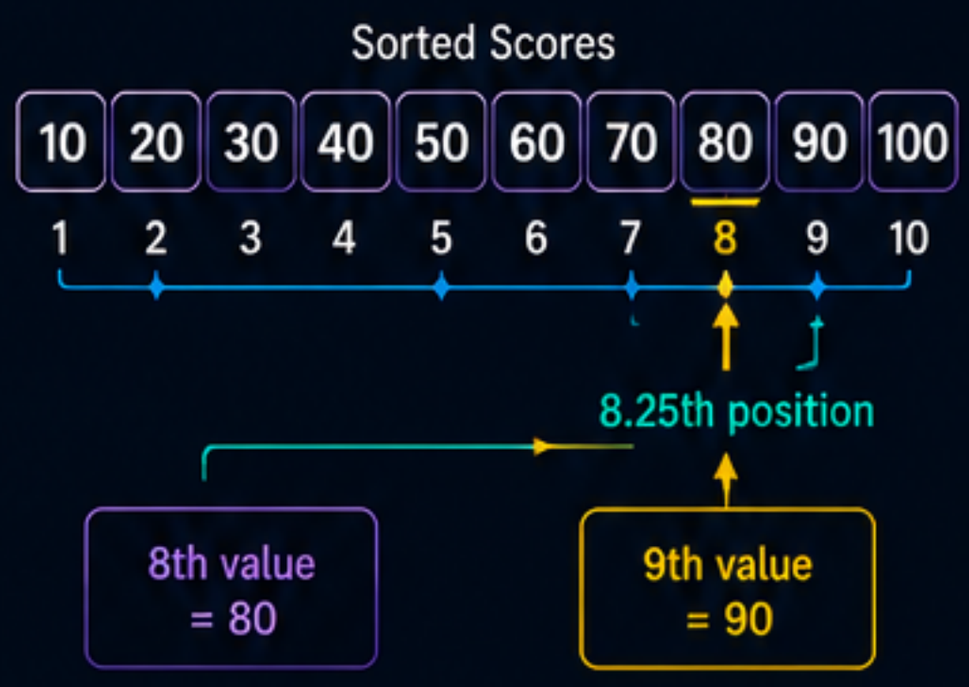
Step 3: Locate the value

8.25^{th} position lies between 8^{th} value (80) and 9^{th} value (90).

So, P75 = $80 + 0.25 \times (90 - 80)$

$= 80 + 2.5 = 82.5$

P75 = 82.5



Interpretation

P75 = 82.5 means 75% of students scored 82.5 or less, and 25% scored more than 82.5.

Business Use Case

In HR, P90 salary helps identify top 10% high earners. In sales, P75 helps identify customers who contribute more than average.

Key Takeaway

Percentiles help us compare a value with the whole dataset.

QUICK SUMMARY – COMMON PERCENTILES

P25	(25th percentile)	1st Quartile (Q1)	25% of data below this value
P50	(50th percentile)	Median (Q2)	50% of data below this value
P75	(75th percentile)	3rd Quartile (Q3)	75% of data below this value
P90	(90th percentile)	90th Percentile	90% of data below this value
P95	(95th percentile)	95th Percentile	95% of data below this value

PERCENTILES vs QUARTILES

Concept	Divides Into	Key Points
Percentiles	100 parts	More detailed Example: P90
Quartiles	4 parts	Simpler summary Example: Q1, Q2, Q3

Quick Check

Scores (sorted): 5, 15, 25, 35, 45, 55, 65, 75, 85, 95

Find P25, P50 (Median) and P75.

Answer

P25 = 20
P50 = 50
P75 = 80

Pro Tip

Always sort the data before calculating percentiles.

10 INTERQUARTILE RANGE (IQR)

Measures of Spread

Definition

The Interquartile Range (IQR) is the range of the middle 50% of the dataset.

IQR = Q3 – Q1

In Simple Words

It tells us how spread out the middle half of the data is.

It is less affected by outliers than the overall range.

Why Useful?

- Shows variability in the middle 50% of data
- Robust measure
- Used in box plots and outlier detection

EXAMPLE – Monthly Expenses (in ₹)

Sorted Data:

10k | 15k | 18k | 20k | 22k | 25k | 28k | 30k | 35k | 40k

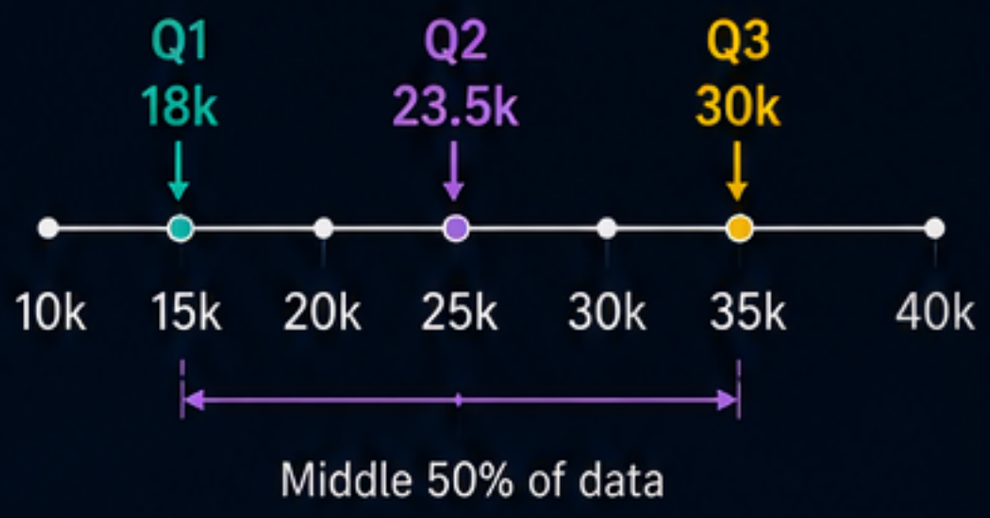
Step 1: Find Quartiles

Q1 (25th Percentile)	Q2 (Median)	Q3 (75th Percentile)
18k	23.5k	30k

Step 2: Calculate IQR

IQR = Q3 – Q1 = 30k – 18k = 12k

Visual Representation



Business Use Case

A finance team uses IQR to understand the spread of monthly expenses without being affected by extreme high or low values.

Key Takeaway

IQR = 12k means the middle 50% of expenses lie within a range of ₹12,000.

11 FIVE NUMBER SUMMARY (For Box Plot)

1. Minimum — The smallest value in the dataset.

2. Q1 (25th %ile) — 25% of values are less than or equal to Q1.

3. Median (Q2) — The middle value. 50% of values are on each side.

4. Q3 (75th %ile) — 75% of values are less than or equal to Q3.

5. Maximum — The largest value in the dataset.

EXAMPLE – Test Scores

Sorted Data:

12 | 18 | 22 | 25 | 28 | 30 | 32 | 35 | 40 | 45

Measure	Value
Minimum	12
Q1	22
Median (Q2)	29
Q3	35
Maximum	45

Box Plot Representation

Business Use Case

Teachers use five number summary and box plots to analyse student performance and detect outliers (low or extremely high scorers).

Key Takeaway

It gives a complete picture of the data's spread, center and extremes.

Quick Check

For data: 5, 8, 12, 18, 20, 25, 30

Find the five number summary.

Answer

Min = 5, Q1 = 10, Median = 19, Q3 = 25, Max = 30

12 MEASURES OF CENTRAL TENDENCY (SUMMARY)

The three main measures that describe the center or typical value of a dataset.

MEAN (Arithmetic Mean)

Average of all values.
Uses every value in the dataset.

$$\text{Mean} = \frac{\sum x_i}{n}$$

MEDIAN

Middle value when data is sorted.
Not affected by extreme values.

Median = Middle value

MODE

Most frequent value (occurs the most times).

Mode = Most frequent value

EXAMPLE – Student Marks

Data (marks out of 100): 45, 50, 55, 55, 55, 60, 65, 70, 75, 80

1. MEAN

Add all values and divide by count.
Sum = 45+50+55+55+60+65+70+75+80 = 555

Mean = 555 / 9 = **61.67**

2. MEDIAN

Data is already sorted.
Middle position = (9+1)/2 = 5th value

45, 50, 55, 55, **60**, 65, 70, 75, 80

Median = 60

3. MODE

55 appears 2 times.
All other values appear once.

Mode = 55

Business Use Case

A merchant wants to know the typical sales.

- Mean = overall average sales
- Median = middle sale value (good for skewed data)
- Mode = most common sale amount

Key Takeaway

- Mean uses all values and is sensitive to outliers.
- Median is the middle value and is robust.
- Mode is the most common value (may be more than one).

13 MEASURES OF DISPERSION (SUMMARY)

These measures tell us how spread out the data is.

Measure	What it Indicates	Formula (Sample)	Affected by Outliers?	Best Used When
Range	Spread between highest and lowest value	Max – Min	Yes	Quick idea of spread (simple data)
σ^2 Variance	Average of squared deviations from mean	$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$	Yes	Statistical analysis, model building
$\sum$ Standard Deviation (SD)	Average distance of values from mean	$s = \sqrt{s^2}$	Yes	Most widely used measure of spread
$\lvert\bar{x}\rvert$ Mean Absolute Deviation (MAD)	Average of absolute deviations from mean	$\frac{\sum \lvert x_i - \bar{x}\rvert}{n}$	Less	When outliers are present
CV% Coefficient of Variation (CV)	Relative comparison of variability	$\frac{s}{\bar{x}} \times 100$	Yes	Comparing datasets with different units

Side-by-Side Example

Data: 2, 4, 6, 8, 10

Measure	Value
Range	10 – 2 = 8
Variance (s^2)	10
Standard Deviation (s)	3.16
MAD	2.4
CV (%)	52.67%

Why Use These?

- Central tendency tells us the center.
- Dispersion tells us the reliability of that center.
- Low spread = data is consistent.
- High spread = data is variable.

Real World Impact

- Quality control: Low SD means stable product quality.
- Finance: Low risk investments have lower volatility (SD).
- Marketing: Understand customer spend patterns (range, median, SD).

Quick Check

For data: 3, 7, 9, 9, 12
Find Mean, Median, Mode and Range.

Answer

Mean = 8, Median = 9, Mode = 9, Range = 12 – 3 = 9

14 BOX PLOT (FIVE NUMBER SUMMARY)

A box plot is a visual way to display the five number summary and understand the spread of data.

What is it?

A box plot shows the minimum, Q1, median, Q3 and maximum on a number line.

Why is it useful?

- Quickly shows spread, center and outliers
- Easy to compare multiple datasets
- Works well for skewed data

How to read?

- Box = IQR (middle 50%)
- Line inside box = Median
- Whiskers = Min to Max (without outliers)
- Points beyond whiskers = Outliers

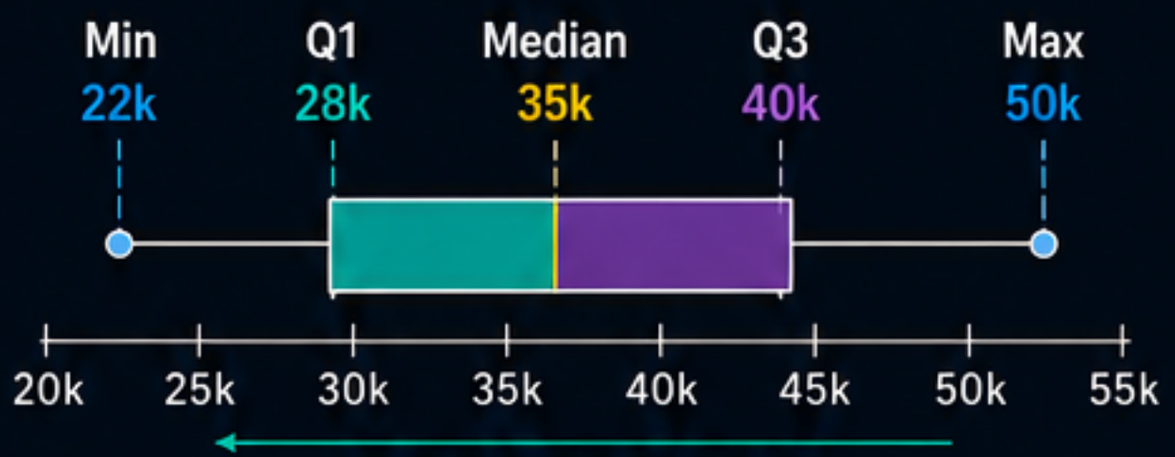
EXAMPLE – Monthly Salaries (in ₹)

Sorted Data:

22k | 25k | 28k | 30k | 32k | 35k | 38k | 40k | 45k | 50k

Measure	Value
Minimum	22k
Q1 (25th Percentile)	28k
Median (Q2)	35k
Q3 (75th Percentile)	40k
Maximum	50k
IQR (Q3 – Q1)	12k

Box Plot Representation



IQR = 40k – 28k = 12k
(Middle 50% of data)

Business Use Case

HR teams use box plots to compare salary distributions across departments and detect outliers (very high or very low salaries).

Key Takeaway

Box plots provide a clear picture of data distribution, center, spread and outliers in one simple chart.

15 HISTOGRAM

Definition

A histogram is a type of bar chart that shows the frequency distribution of continuous data using contiguous bars.

Key Points

- X-axis: Continuous intervals
- Y-axis: Frequency
- Bars touch each other (no gaps)
- Width of bars = class interval

When to use?

- To understand the shape of data distribution
- To identify skewness, center and spread
- Common in quality control, finance, analytics

EXAMPLE – Order Amounts (₹)

Raw Data (in ₹):

120, 150, 200, 220, 250, 270, 300, 320, 350, 360, 410, 450, 480, 520, 540, 580, 610, 650, 700, 720, 780, 800, 850, 900, 950, 980, 1000

Step 1: Create Class Intervals
(Class width = 100)

Step 2: Count Frequencies

Order Amount (₹)	Frequency
100 – 199	2
200 – 299	5
300 – 399	5
400 – 499	4
500 – 599	4
600 – 699	3
700 – 799	3
800 – 899	2
900 – 999	2
Total	30

Histogram

Business Use Case

E-commerce companies use histograms to see how orders are distributed across different value ranges.

Insight

Most orders fall between ₹200 – ₹399.
Very high order values (₹800+) are rare.

Quick Check

For the data: 22, 25, 28, 30, 32, 35, 38, 40, 49, 45, 50
Find Min, Q1, Median, Q3, Max and IQR.

Answer

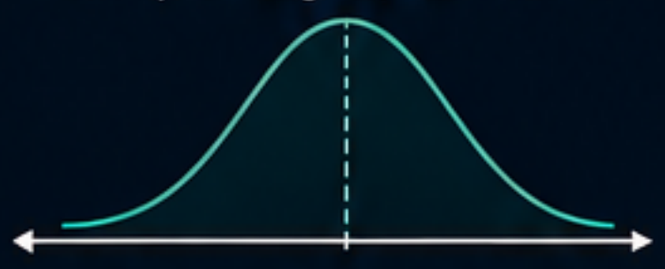
Min = 22, Q1 = 28, Median = 35, Q3 = 40, Max = 50, IQR = 12

16 TYPES OF DISTRIBUTIONS (SHAPE OF DATA)

Understanding the shape helps in better interpretation and decision making.

1. SYMMETRICAL DISTRIBUTION

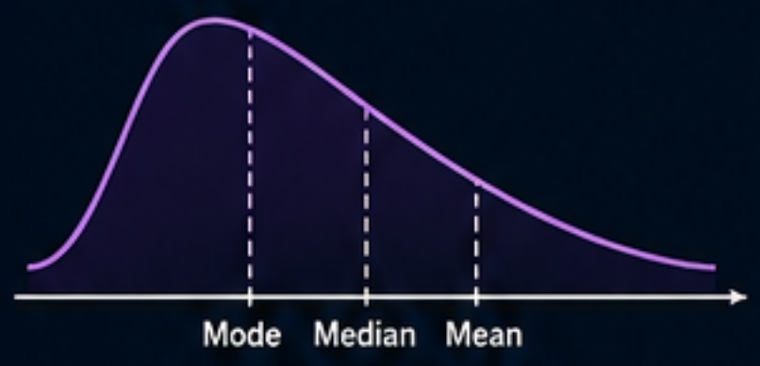
- Data is evenly distributed around the center.
- Mean = Median = Mode
- Example: Heights, Test scores



Mean = Median = Mode

2. RIGHT SKEWED DISTRIBUTION

- Long tail on the right side.
- Mean > Median > Mode
- Example: Income, House prices



Mean > Median > Mode

3. LEFT SKEWED DISTRIBUTION

- Long tail on the left side.
- Mean < Median < Mode
- Example: Age at retirement

Mean < Median < Mode

EXAMPLE – Exam Scores Distribution

Raw Scores (out of 100):

32	45	50	52	55	58	60	65	68	70	72
75	78	80	82	85	88	90	91	92	95	

Business Use Case

Retailers analyze income (right skewed) to plan premium product segments for top income customers.

Key Insight

Knowing the shape helps in selecting the right statistical methods and predicting outcomes better.

Pro Tip

Always visualize your data with histograms or box plots to understand its shape before analysis.

MODULE 2 – DESCRIPTIVE STATISTICS: SUMMARY

1	Frequency Distribution	Shows how often each value occurs in data.	9	Percentiles	Divide data into 100 parts (e.g., P25, P50, P75).
2	Frequency Table	Organized table of values with their frequencies.	10	Quartiles (IQR)	Divide data into 4 parts. IQR = Q3 – Q1
3	Mean (Arithmetic Mean)	Average of all values.	11	Five Number Summary	Min, Q1, Median, Q3, Max. Great for box plots.
4	Median	Middle value when data is sorted.	12	Measures of Central Tendency	Mean, Median, Mode – describe the center.
5	Mode	Most frequently occurring value.	13	Measures of Dispersion	Range, Variance, SD, MAD, CV, IQR – describe the spread.
6	Geometric Mean	Used for growth rates and compound changes.	14	Box Plot	Visual tool to show 5-number summary and outliers.
7	Weighted Mean	Average where some values have more importance.	15	Histogram	Shows the distribution shape of continuous data.
8	Harmonic Mean	Used for rates and ratios.	16	Types of Distributions	Symmetrical, Right Skewed, Left Skewed.

Why This Matters?

Descriptive statistics helps us turn raw data into meaningful insights, make informed decisions and communicate findings clearly.

Remember

"Good decisions start with understanding the data."

Up Next in Module 3

- Introduction to Inferential Statistics
- Probability Basics
- Sampling & Sampling Distributions

You did it! 🎉

You've completed Module 2 – Descriptive Statistics.
Strong foundation for advanced data analysis!

Keep Learning!